



A NEW APPROACH TO EQUILIBRIUM PROBLEMS VIA REGULARIZED EXTRAGRADIENT-TYPE DYNAMICS

SUPPLEMENTARY MATERIAL: DETAILED PROOF

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Regularized Dynamics of Extragradient-Type for Equilibrium Problems

Let C be a nonempty, closed and convex set in a real Hilbert space H , $f : H \times H \rightarrow \mathbb{R}$ be an equilibrium bifunction, i.e., $f(x, x) = 0$ for all $x \in H$. Consider the following equilibrium problem

$$\text{Find } x^* \in C \text{ such that } f(x^*, y) \geq 0 \quad \forall y \in C. \quad (1)$$

Denote by $Sol(f, C)$ the solution set of (1). In what follows, we assume that $Sol(f, C)$ is not empty. We focus on the following dynamics associated with (1):

$$\begin{cases} z(t) = \text{prox}_{\lambda(t)(f(x(t), \cdot) + \alpha(t)g(x(t), \cdot))}(x(t)), \\ y(t) = \text{prox}_{\lambda(t)(f(x(t), \cdot) + \alpha(t)g(z(t), \cdot))}(z(t)), \\ \dot{x}(t) + x(t) = y(t), \end{cases} \quad (2)$$

or equivalently,

$$\begin{cases} z(t) = \text{argmin}\{\lambda(t)f(x(t), u) + \lambda(t)\alpha(t)g(x(t), u) + \frac{1}{2}\|u - x(t)\|^2 : u \in C\}, \\ y(t) = \text{argmin}\{\lambda(t)f(x(t), v) + \lambda(t)\alpha(t)g(z(t), v) + \frac{1}{2}\|v - z(t)\|^2 : v \in C\}, \\ \dot{x}(t) + x(t) = y(t). \end{cases} \quad (3)$$

Discretizing (2) by approximating $\dot{x}(t_k) \approx \frac{x(t_{k+1}) - x(t_k)}{\tau_k}$ leads to several regularized extragradient-type algorithms with relaxation parameters, namely
REA1 (Regularized Extragradient-Type Algorithm 1)

$$\begin{cases} z^k = \text{prox}_{\lambda_k(f(x^k, \cdot) + \alpha_k g(x^k, \cdot))}(x^k), \\ y^k = \text{prox}_{\lambda_k(f(x^k, \cdot) + \alpha_k g(z^k, \cdot))}(z^k), \\ x^{k+1} = (1 - \tau_k)x^k + \tau_k y^k. \end{cases} \quad (4)$$

REA2 (Regularized Extragradient-Type Algorithm 2)

$$\begin{cases} z^k = \text{prox}_{\lambda_k(f(x^k, \cdot) + \alpha_k g(x^k, \cdot))}(x^k), \\ y^k = \text{prox}_{\lambda_{k+1}(f(x^k, \cdot) + \alpha_{k+1} g(z^k, \cdot))}(z^k), \\ x^{k+1} = (1 - \tau_k)x^k + \tau_k y^k. \end{cases} \quad (5)$$

The following issues should be solved:

- i. The existence and uniqueness of the strong global solution $x(t)$ to (2).
- ii. The strong convergence of $x(t)$ to a special chosen solution $x^\dagger$ of the equilibrium problem (1).
- iii. The strong convergence of regularized extragradient-type algorithms REA1, REA2 to the solution $x^\dagger$.
- iv. Convergence rate of the REA1, REA2 under the strong monotonicity of the bifunction $f(x, y)$.

v. Numerical experiments.

Answer for iii. We first bring out the following hypotheses for a bifunction $f : C \times C \rightarrow \mathbb{R}$:

(H1) f is monotone and $f(u, u) = 0$ for all $u \in C$.

(H1') f is γ -strongly monotone.

(H2) $f(u, \cdot)$ is convex and weakly lower semi-continuous, $f(\cdot, v)$ is weakly upper semi-continuous.

(H3) The solution set $Sol(f, C)$ is nonempty.

Now, we take algorithm REA1 into consideration.

Statement 1. Assume that hypotheses (H1), (H2), (H3) hold for the bifunction f ; and (H1'), (H2) hold for the (uniformly bounded) bifunction g . Moreover, f satisfies the L -Lipschitz type condition and g satisfies the K -Lipschitz type condition. Besides, the sequences $\{\tau_k\} \subset [a, b] \subset (0, 1)$, $\{\lambda_k\}$, $\{\alpha_k\} \subset (0, +\infty)$ satisfy

$$\sum_{k=0}^{+\infty} \lambda_k \alpha_k = +\infty; \quad \lim_{k \rightarrow \infty} \alpha_k = 0; \quad \lim_{k \rightarrow \infty} \frac{\lambda_k}{\alpha_k} = 0; \quad \lim_{k \rightarrow \infty} \frac{|\alpha_k - \alpha_{k+1}|}{\alpha_k^2} = 0. \quad (6)$$

Then, the sequence $\{x^k\}$ generated by algorithm REA1 converges strongly to a solution $x^\dagger$ of problem 1, where $x^\dagger \in Sol(g, Sol(f, C))$.

Proof of statement 1. For each $\alpha > 0$, let x_α denote the unique solution of a regularized problem

$$\text{Find } x^* \in C \text{ such that } f(x^*, y) + \alpha g(x^*, y) \geq 0 \quad \forall y \in C. \quad (7)$$

It is sufficient to prove that $x^k - x_{\alpha_k}$ as $k \rightarrow \infty$ since we already have $x_{\alpha_k} \rightarrow x^\dagger$. We see that

$$\begin{aligned} \|x^{k+1} - x_{\alpha_{k+1}}\|^2 &= \|(1 - \tau_k)x^k + \tau_k y^k - x_{\alpha_{k+1}}\|^2 \\ &= \|(x^k - x_{\alpha_k}) + \tau_k(y^k - x^k) + (x_{\alpha_k} - x_{\alpha_{k+1}})\|^2 \\ &= \|x^k - x_{\alpha_k}\|^2 + \tau_k^2 \|y^k - x^k\|^2 + \|x_{\alpha_k} - x_{\alpha_{k+1}}\|^2 \\ &\quad + 2\langle x^k - x_{\alpha_k}, x_{\alpha_k} - x_{\alpha_{k+1}} \rangle + 2\tau_k \langle y^k - x^k, x_{\alpha_k} - x_{\alpha_{k+1}} \rangle + 2\tau_k \langle x^k - x_{\alpha_k}, y^k - x^k \rangle \\ &\leq \|x^k - x_{\alpha_k}\|^2 + \tau_k^2 \|x^k - y^k\|^2 + M^2 \frac{|\alpha_k - \alpha_{k+1}|^2}{\alpha_k^2} + 2M \frac{|\alpha_k - \alpha_{k+1}|}{\alpha_k} \|x^k - x_{\alpha_k}\| \\ &\quad + 2\tau_k M \frac{|\alpha_k - \alpha_{k+1}|}{\alpha_k} \|x^k - y^k\| + 2\tau_k \langle x^k - x_{\alpha_k}, y^k - x^k \rangle. \end{aligned} \quad (8)$$

Next, we have

$$\begin{aligned} \langle x^k - x_{\alpha_k}, y^k - x^k \rangle &= \langle y^k - x_{\alpha_k}, y^k - x^k \rangle - \|x^k - y^k\|^2 \\ &= \langle y^k - x_{\alpha_k}, y^k - z^k \rangle + \langle z^k - x_{\alpha_k}, z^k - x^k \rangle + \langle y^k - z^k, z^k - x^k \rangle - \|x^k - y^k\|^2. \end{aligned} \quad (9)$$

Because $y^k = \text{prox}_{\lambda_k(f(x^k, \cdot) + \alpha_k g(z^k, \cdot))}(z^k)$, it follows from lemma 1 in [1] that for all $v \in C$,

$$\langle y^k - v, y^k - z^k \rangle \leq \lambda_k [f(x^k, v) + \alpha_k g(z^k, v)] - \lambda_k [f(x^k, y^k) + \alpha_k g(z^k, y^k)].$$

We substitute $v = x_{\alpha_k}$ into this inequality and re-organize to obtain

$$\langle y^k - x_{\alpha_k}, y^k - z^k \rangle \leq \lambda_k [f(x^k, x_{\alpha_k}) - f(x^k, y^k)] + \lambda_k \alpha_k [g(z^k, x_{\alpha_k}) - g(z^k, y^k)]. \quad (10)$$

It follows from the Lipschitz type condition of f and g that

$$f(x^k, x_{\alpha_k}) - f(x^k, y^k) \leq L\|x^k - y^k\|\|x_{\alpha_k} - y^k\| + f(y^k, x_{\alpha_k}) \quad (11)$$

and

$$g(z^k, x_{\alpha_k}) - g(z^k, y^k) \leq K\|z^k - y^k\|\|x_{\alpha_k} - y^k\| + g(y^k, x_{\alpha_k}). \quad (12)$$

We combine (10), (11) and (12) to get

$$\begin{aligned} \langle y^k - x_{\alpha_k}, y^k - z^k \rangle &\leq \lambda_k L\|x^k - y^k\|\|x_{\alpha_k} - y^k\| + \lambda_k \alpha_k K\|z^k - y^k\|\|x_{\alpha_k} - y^k\| \\ &\quad + \lambda_k [f(y^k, x_{\alpha_k}) + \alpha_k g(y^k, x_{\alpha_k})]. \end{aligned} \quad (13)$$

Since x_{α_k} is solution of a problem (7), we have

$$f(x_{\alpha_k}, y^k) + \alpha_k g(x_{\alpha_k}, y^k) \geq 0. \quad (14)$$

The monotonicity of f together with the γ -strong monotonicity of g implies that

$$[f(x_{\alpha_k}, y^k) + f(y^k, x_{\alpha_k})] + \alpha_k [g(x_{\alpha_k}, y^k) + g(y^k, x_{\alpha_k})] \leq -\gamma \alpha_k \|x_{\alpha_k} - y^k\|^2,$$

or equivalently,

$$[f(x_{\alpha_k}, y^k) + \alpha_k g(x_{\alpha_k}, y^k)] + [f(y^k, x_{\alpha_k}) + \alpha_k g(y^k, x_{\alpha_k})] \leq -\gamma \alpha_k \|x_{\alpha_k} - y^k\|^2. \quad (15)$$

From (14) and (15), we get

$$f(y^k, x_{\alpha_k}) + \alpha_k g(y^k, x_{\alpha_k}) \leq -\gamma \alpha_k \|x_{\alpha_k} - y^k\|^2. \quad (16)$$

We combine (13) and (16) to reach

$$\begin{aligned} \langle y^k - x_{\alpha_k}, y^k - z^k \rangle &\leq \lambda_k L\|x^k - y^k\|\|x_{\alpha_k} - y^k\| + \lambda_k \alpha_k K\|z^k - y^k\|\|x_{\alpha_k} - y^k\| \\ &\quad - \gamma \lambda_k \alpha_k \|x_{\alpha_k} - y^k\|^2. \end{aligned} \quad (17)$$

Next, because $z^k = \text{prox}_{\lambda_k(f(x^k, \cdot) + \alpha_k g(x^k, \cdot))}(x^k)$, we once again use lemma 1 in [1] to get

$$\langle z^k - v, z^k - x^k \rangle \leq \lambda_k [f(x^k, v) + \alpha_k g(x^k, v)] - \lambda_k [f(x^k, z^k) + \alpha_k g(x^k, z^k)]$$

for all $v \in C$. We then substitute $v = x_{\alpha_k}$ and repeat the transformations from (10) to (17) to see that

$$\begin{aligned} \langle z^k - x_{\alpha_k}, z^k - x^k \rangle &\leq \lambda_k L\|x^k - z^k\|\|x_{\alpha_k} - z^k\| + \lambda_k \alpha_k K\|x^k - z^k\|\|x_{\alpha_k} - z^k\| \\ &\quad - \gamma \lambda_k \alpha_k \|x_{\alpha_k} - z^k\|^2 \\ &= \lambda_k (L + \alpha_k K)\|x^k - z^k\|\|x_{\alpha_k} - z^k\| - \gamma \lambda_k \alpha_k \|x_{\alpha_k} - z^k\|^2 \end{aligned} \quad (18)$$

On the other hand, it is easy to see that

$$\begin{aligned} \langle y^k - z^k, z^k - x^k \rangle &= \langle y^k - z^k, y^k - x^k \rangle - \|y^k - z^k\|^2 \\ &\leq \frac{1}{2} (\|y^k - z^k\|^2 + \|y^k - x^k\|^2) - \|y^k - z^k\|^2 \\ &= \frac{1}{2} \|y^k - x^k\|^2 - \frac{1}{2} \|y^k - z^k\|^2. \end{aligned} \quad (19)$$

We infer from (9), (17), (18) and (19) that

$$\begin{aligned}\langle x^k - x_{\alpha_k}, y^k - x^k \rangle &\leq \lambda_k L \|x^k - y^k\| \|x_{\alpha_k} - y^k\| + \lambda_k \alpha_k K \|z^k - y^k\| \|x_{\alpha_k} - y^k\| \\ &\quad - \gamma \lambda_k \alpha_k \|x_{\alpha_k} - y^k\|^2 + \lambda_k (L + \alpha_k K) \|x^k - z^k\| \|x_{\alpha_k} - z^k\| \\ &\quad - \gamma \lambda_k \alpha_k \|x_{\alpha_k} - z^k\|^2 - \frac{1}{2} \|y^k - x^k\|^2 - \frac{1}{2} \|y^k - z^k\|^2.\end{aligned}\quad (20)$$

We combine (8) and (20) to get

$$\begin{aligned}\|x^{k+1} - x_{\alpha_{k+1}}\|^2 &\leq \|x^k - x_{\alpha_k}\|^2 + (\tau_k^2 - \tau_k) \|x^k - y^k\|^2 + \frac{M^2 |\alpha_k - \alpha_{k+1}|^2}{\alpha_k^2} + 2M \frac{|\alpha_k - \alpha_{k+1}|}{\alpha_k} \|x^k - x_{\alpha_k}\| \\ &\quad + 2\tau_k M \frac{|\alpha_k - \alpha_{k+1}|}{\alpha_k} \|x^k - y^k\| + 2\tau_k \lambda_k L \|x^k - y^k\| \|x_{\alpha_k} - y^k\| \\ &\quad + 2\tau_k \lambda_k \alpha_k K \|z^k - y^k\| \|x_{\alpha_k} - y^k\| + 2\tau_k \lambda_k (L + \alpha_k K) \|x^k - z^k\| \|x_{\alpha_k} - z^k\| \\ &\quad - 2\gamma \tau_k \lambda_k \alpha_k \|x_{\alpha_k} - y^k\|^2 - 2\gamma \tau_k \lambda_k \alpha_k \|x_{\alpha_k} - z^k\|^2 - \tau_k \|y^k - z^k\|^2.\end{aligned}\quad (21)$$

Since $\tau_k \in [a, b] \subset (0, 1)$, we can choose a number $c > 0$ such that

$$\tau_k^2 - \tau_k \leq -4c \text{ for every } k.$$

Indeed, the above inequality holds for $c = \max\{\frac{1}{4}(a - a^2), \frac{1}{4}(b - b^2)\}$. Thus, we have

$$(\tau_k^2 - \tau_k) \|x^k - y^k\|^2 \leq -4c \|x^k - y^k\|^2. \quad (22)$$

Next, we also choose a positive number $c_1 > 0$ such that $c_1 \leq \min\{\frac{a}{2}, c\}$. Using Cauchy-Schwartz's inequality, we see that

$$2\tau_k \lambda_k L \|x^k - y^k\| \|x_{\alpha_k} - y^k\| \leq c \|x^k - y^k\|^2 + \frac{\tau_k^2 \lambda_k^2 L^2}{c} \|x_{\alpha_k} - y^k\|^2, \quad (23)$$

$$2\tau_k \lambda_k \alpha_k K \|z^k - y^k\| \|x_{\alpha_k} - y^k\| \leq \frac{\tau_k}{2} \|z^k - y^k\|^2 + 2\tau_k \lambda_k^2 \alpha_k^2 K^2 \|x_{\alpha_k} - y^k\|^2, \quad (24)$$

and

$$\begin{aligned}2\tau_k \lambda_k (L + \alpha_k K) \|x^k - z^k\| \|x_{\alpha_k} - z^k\| &\leq \frac{c_1}{2} \|x^k - z^k\|^2 + \frac{2\tau_k^2 \lambda_k^2 (L + \alpha_k K)^2}{c_1} \|x_{\alpha_k} - z^k\|^2 \\ &\leq c_1 \|x^k - y^k\|^2 + c_1 \|y^k - z^k\|^2 + \frac{2\tau_k^2 \lambda_k^2 (L + \alpha_k K)^2}{c_1} \|x_{\alpha_k} - z^k\|^2\end{aligned}\quad (25)$$

where we used the fact that $\frac{1}{2} \|x^k - z^k\|^2 \leq \|x^k - y^k\|^2 + \|y^k - z^k\|^2$ in (25). For simplicity, we set

$$\begin{cases} \text{SH1}_k = \left(\frac{\tau_k^2 \lambda_k^2 L^2}{c} + 2\tau_k \lambda_k^2 \alpha_k^2 K^2 - \gamma \tau_k \lambda_k \alpha_k \right) \|x_{\alpha_k} - y^k\|^2, \\ \text{SH2}_k = \left(\frac{2\tau_k^2 \lambda_k^2 (L + \alpha_k K)^2}{c_1} - 2\gamma \tau_k \lambda_k \alpha_k \right) \|x_{\alpha_k} - z^k\|^2, \\ \text{SH3}_k = \left(c_1 - \frac{\tau_k}{2} \right) \|y^k - z^k\|^2. \end{cases}$$

It follows from (21), (22), (23), (24) and (25) that

$$\begin{aligned} \|x^{k+1} - x_{\alpha_{k+1}}\|^2 &\leq \|x^k - x_{\alpha_k}\|^2 - 2c\|x^k - y^k\|^2 + \frac{M^2|\alpha_k - \alpha_{k+1}|^2}{\alpha_k^2} + 2M\frac{|\alpha_k - \alpha_{k+1}|}{\alpha_k}\|x^k - x_{\alpha_k}\| \\ &\quad + 2\tau_k M\frac{|\alpha_k - \alpha_{k+1}|}{\alpha_k}\|x^k - y^k\| + \text{SH1}_k + \text{SH2}_k + \text{SH3}_k - \gamma\tau_k\lambda_k\alpha_k\|x_{\alpha_k} - y^k\|^2. \end{aligned} \quad (26)$$

It is easy to see that

$$\|x_{\alpha_k} - y^k\|^2 \geq \frac{1}{2}\|x^k - x_{\alpha_k}\|^2 - \|x^k - y^k\|^2. \quad (27)$$

The relation (26) together with (27) and Cauchy-Schwartz's inequality leads to

$$\begin{aligned} \|x^{k+1} - x_{\alpha_{k+1}}\|^2 &\leq \|x^k - x_{\alpha_k}\|^2 - 2c\|x^k - y^k\|^2 + \frac{M^2|\alpha_k - \alpha_{k+1}|^2}{\alpha_k^2} \\ &\quad + \left(\frac{\gamma\tau_k\lambda_k\alpha_k}{4}\|x^k - x_{\alpha_k}\|^2 + \frac{4M^2|\alpha_k - \alpha_{k+1}|^2}{\gamma\tau_k\lambda_k\alpha_k^3}\right) + \left(c\|x^k - y^k\|^2 + \frac{\tau_k^2 M^2|\alpha_k - \alpha_{k+1}|^2}{c\alpha_k^2}\right) \\ &\quad + \text{SH1}_k + \text{SH2}_k + \text{SH3}_k - \gamma\tau_k\lambda_k\alpha_k\left(\frac{1}{2}\|x^k - x_{\alpha_k}\|^2 - \|x^k - y^k\|^2\right). \\ &= \left(1 - \frac{\gamma\tau_k\lambda_k\alpha_k}{4}\right)\|x^k - x_{\alpha_k}\|^2 + (\gamma\tau_k\lambda_k\alpha_k - c)\|x^k - y^k\|^2 \\ &\quad + \left(\frac{\tau_k^2}{c} + 1\right)\frac{M^2|\alpha_k - \alpha_{k+1}|^2}{\alpha_k^2} + \frac{4M^2|\alpha_k - \alpha_{k+1}|^2}{\gamma\tau_k\lambda_k\alpha_k^3} + \text{SH1}_k + \text{SH2}_k + \text{SH3}_k. \end{aligned} \quad (28)$$

According to the conditions that $\{h_k\}$ is bounded, $\lim \alpha_k = 0$ and $\lim_{\alpha_k} \frac{\lambda_k}{\alpha_k} = 0$ as $k \rightarrow \infty$, without loss of generality we may assume that

$$\begin{cases} \frac{\tau_k^2\lambda_k^2L^2}{c} + 2\tau_k\lambda_k^2\alpha_k^2K^2 - \gamma\tau_k\lambda_k\alpha_k \leq 0, \\ \frac{2\tau_k^2\lambda_k^2(L + \alpha_kK)^2}{c_1} - 2\gamma\tau_k\lambda_k\alpha_k \leq 0, \\ \gamma\tau_k\lambda_k\alpha_k - c \leq 0. \end{cases} \quad (29)$$

It follows from (29) and the definition of c_1 that SH1_k , SH2_k , SH3_k and $(\gamma\tau_k\lambda_k\alpha_k - c)\|x^k - y^k\|^2$ are all non-positive. Therefore, we imply from (28) that

$$\|x^{k+1} - x_{\alpha_{k+1}}\|^2 \leq \left(1 - \frac{\gamma\tau_k\lambda_k\alpha_k}{4}\right)\|x^k - x_{\alpha_k}\|^2 + \left(\frac{\tau_k^2}{c} + 1\right)\frac{M^2|\alpha_k - \alpha_{k+1}|^2}{\alpha_k^2} + \frac{4M^2|\alpha_k - \alpha_{k+1}|^2}{\gamma\tau_k\lambda_k\alpha_k^3}. \quad (30)$$

We denote

$$\begin{aligned}\psi_k &= \|x^k - x_{\alpha_k}\|^2; \\ p_k &= \frac{\gamma\tau_k\lambda_k\alpha_k}{4}; \\ q_k &= \left(\frac{\tau_k^2}{c} + 1\right) \frac{M^2|\alpha_k - \alpha_{k+1}|^2}{\alpha_k^2} + \frac{4M^2|\alpha_k - \alpha_{k+1}|^2}{\gamma\tau_k\lambda_k\alpha_k^3}.\end{aligned}$$

From (6), we have $\lambda_k\alpha_k \rightarrow 0$ as $k \rightarrow \infty$ and $\sum_{k=0}^{\infty} \lambda_k\alpha_k = +\infty$. Thus, we see that

$$\lim_{k \rightarrow \infty} p_k = 0 \quad \text{and} \quad \sum_{k=0}^{\infty} p_k = +\infty \quad (31)$$

and it can be assumed without loss of generality that

$$p_k \in (0, 1) \quad (32)$$

for all k . Furthermore, using the condition that

$$\lim_{k \rightarrow \infty} \frac{|\alpha_k - \alpha_{k+1}|}{\lambda_k\alpha_k^2} = 0,$$

we get

$$\frac{M^2|\alpha_k - \alpha_{k+1}|^2}{\alpha_k^2} \cdot \frac{1}{p_k} = \frac{4M^2}{\gamma\tau_k} \cdot \left(\frac{|\alpha_k - \alpha_{k+1}|}{\lambda_k\alpha_k^2}\right)^2 \cdot \lambda_k\alpha_k \rightarrow 0 \text{ as } k \rightarrow \infty, \quad (33)$$

and

$$\frac{4M^2|\alpha_k - \alpha_{k+1}|^2}{\gamma\tau_k\lambda_k\alpha_k^3} \cdot \frac{1}{p_k} = \frac{16M^2}{\gamma^2\tau_k^2} \cdot \left(\frac{|\alpha_k - \alpha_{k+1}|}{\lambda_k\alpha_k^2}\right)^2 \rightarrow 0 \text{ as } k \rightarrow \infty. \quad (34)$$

We combine (33), (34) and the fact that the sequence $\{\tau_k\}$ is bounded to imply that

$$\lim_{k \rightarrow \infty} \frac{q_k}{p_k} = 0. \quad (35)$$

Consequently, we have

$$\psi_{k+1} \leq (1 - p_k)\psi_k + q_k$$

with the sequences $\{p_k\}$ and $\{q_k\}$ satisfy (31), (32) and (35). Using lemma 2 from [1], we see that $\psi_k \rightarrow 0$ as $k \rightarrow \infty$ or equivalently, $\|x^k - x_{\alpha_k}\|$ converges strongly to 0. The proof is complete.

Statement 2. Assume that all the conditions in statement 1 holds. Then, the sequence $\{x^k\}$ generated by algorithm REA2 converges strongly to a solution $x^\dagger$ of problem 1, where $x^\dagger \in \text{Sol}(g, \text{Sol}(f, C))$.

Proof of statement 2. We repeat the same proof as statement 1 with slight changes. Notice that since $y^k = \text{prox}_{\lambda_{k+1}(f(x^k, \cdot) + \alpha_{k+1}g(z^k, \cdot))}(z^k)$, we have

$$\begin{aligned} \langle y^k - x_{\alpha_k}, y^k - z^k \rangle &\leq \lambda_{k+1}L \|x^k - y^k\| \|x_{\alpha_k} - y^k\| + \lambda_{k+1}\alpha_{k+1}K \|z^k - y^k\| \|x_{\alpha_{k+1}} - y^k\| \\ &\quad - \gamma\lambda_{k+1}\alpha_{k+1} \|x_{\alpha_k} - y^k\|^2. \end{aligned} \quad (36)$$

Using (36) and the same arguments from the proof of statement 1, we finally get

$$\|x^{k+1} - x_{\alpha_{k+1}}\|^2 \leq \left(1 - \frac{\gamma\tau_k\lambda_{k+1}\alpha_{k+1}}{4}\right) \|x^k - x_{\alpha_k}\|^2 + \left(\frac{\tau_k^2}{c} + 1\right) \frac{M^2|\alpha_k - \alpha_{k+1}|^2}{\alpha_k^2} + \frac{4M^2|\alpha_k - \alpha_{k+1}|^2}{\gamma\tau_k\lambda_{k+1}\alpha_{k+1}^3}. \quad (37)$$

We denote

$$\begin{aligned} \psi_k &= \|x^k - x_{\alpha_k}\|^2; \\ p_k &= \frac{\gamma\tau_k\lambda_{k+1}\alpha_{k+1}}{4}; \\ q_k &= \left(\frac{\tau_k^2}{c} + 1\right) \frac{M^2|\alpha_k - \alpha_{k+1}|^2}{\alpha_k^2} + \frac{4M^2|\alpha_k - \alpha_{k+1}|^2}{\gamma\tau_k\lambda_{k+1}\alpha_{k+1}^3}. \end{aligned}$$

Therefore, the relation (37) can be rewritten as

$$\psi_{k+1} \leq (1 - p_k)\psi_k + q_k. \quad (38)$$

It is easy to see that $\lim_{k \rightarrow \infty} p_k = 0$ and $\sum_{k=0}^{\infty} p_k = +\infty$. Hence, it can be assumed without loss of generality that $p_k \in (0, 1)$ for all k .

Next, since the sequences $\{\lambda_k\}$ and $\{\alpha_k\}$ converge, we have

$$\lim_{k \rightarrow \infty} \frac{\lambda_k}{\lambda_{k+1}} = 1 \quad \text{and} \quad \lim_{k \rightarrow \infty} \frac{\alpha_k}{\alpha_{k+1}} = 1. \quad (39)$$

Combine (39) and the condition that

$$\lim_{k \rightarrow \infty} \frac{|\alpha_k - \alpha_{k+1}|}{\lambda_k \alpha_k^2} = 0,$$

we have

$$\frac{M^2|\alpha_k - \alpha_{k+1}|^2}{\alpha_k^2} \cdot \frac{1}{p_k} = \frac{4M^2}{\gamma\tau_k} \cdot \left(\frac{|\alpha_k - \alpha_{k+1}|}{\lambda_k \alpha_k^2}\right)^2 \cdot \frac{\lambda_k}{\lambda_{k+1}} \cdot \frac{\alpha_k}{\alpha_{k+1}} \cdot \lambda_k \alpha_k \rightarrow 0 \text{ as } k \rightarrow \infty, \quad (40)$$

and

$$\frac{4M^2|\alpha_k - \alpha_{k+1}|^2}{\gamma\tau_k\lambda_{k+1}\alpha_{k+1}^3} \cdot \frac{1}{p_k} = \frac{16M^2}{\gamma^2\tau_k^2} \cdot \left(\frac{|\alpha_k - \alpha_{k+1}|}{\lambda_k \alpha_k^2}\right)^2 \cdot \frac{\lambda_k^2}{\lambda_{k+1}^2} \cdot \frac{\alpha_k^4}{\alpha_{k+1}^4} \rightarrow 0 \text{ as } k \rightarrow \infty. \quad (41)$$

The relations (40) and (41) together with the boundedness of $\{\tau_k\}$ infer that

$$\lim_{k \rightarrow \infty} \frac{q_k}{p_k} = 0.$$

We once again apply lemma 2 from [1] to see that $\psi_k \rightarrow 0$ as $k \rightarrow \infty$ or equivalently, $\|x^k - x_{\alpha_k}\|$ converges strongly to 0. The proof is complete.

References

- [1] Hieu, D.V., Quy, P.K., Duong, H.N.: Equilibrium programming and new iterative methods in Hilbert spaces. *Acta Applicandae Mathematicae* **176**(7) (2021)